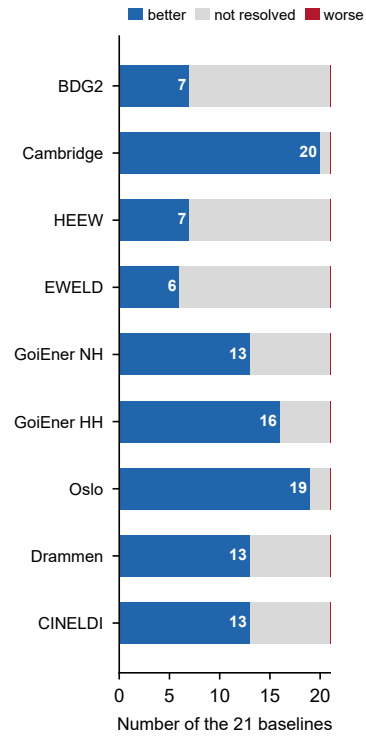
Monthly energy error, 744 × the mean hourly error of the delivered trajectory, on the same test windows (ankyra-lite: its energy readout, the trajectory before the continuation layer, readout rule A). Unit-equal RMS ratio with 95% unit-and-month bootstrap intervals; * resolved in ankyra-lite's favour, (−) against it. BDG2 w/o 3: without the three near-zero meters, where ankyra-lite's error is exactly zero (the point estimates are equal, only the intervals differ). Rows: the 21 baselines (22 forecasters with ankyra-lite), in the order of figure 2. ankyra-lite 0.2.0: carrier recipe Rw (trained from scratch with early stopping on the 93-day validation set, then 8 more epochs with the validation span returned and each training window's loss weighted by the inverse-noise rule; six seeds averaged; figure 1b), the anchoring layer, the state rule and the origin-continuity layer; hourly errors score the delivered trajectory after that layer, energy, level and peak are read before it (readout rule A). Linear-context: the TiDE implementation of the release, whose forecast is a linear map of the load context.

**a ankyra-lite's improvement in monthly energy error over each baseline (%)**

| | BDG2 | BDG2 w/o 3 | Cambridge | HEEW | EWELD | GoiEner NH | GoiEner HH | Oslo | Drammen | CINELDI |
|---|---|---|---|---|---|---|---|---|---|---|
| Ridge (per unit) | +11 | +11 | **+23*** | +2 | +5 | +8 | +2 | **+33*** | -5 | +10 |
| Chronos-2-X | +3 | +3 | **+20*** | -1 | +17 | +12 | **+33*** | **+20*** | +4 | +17 |
| iTransformer-X | **+21*** | **+21*** | +8 | -1 | +5 | +17 | +16 | +10 | +9 | +13 |
| iTransformer | +19 | +19 | **+20*** | +2 | +2 | +10 | **+22*** | **+25*** | +9 | **+15*** |
| PatchTST | +15 | +15 | **+20*** | +5 | -5 | +14 | **+18*** | **+24*** | +9 | +9 |
| TimesFM-X | +18 | +18 | **+26*** | +4 | -2 | **+18*** | **+29*** | **+39*** | **+23*** | **+20*** |
| GBT-T | +11 | +11 | **+23*** | +3 | +25 | **+44*** | **+21*** | +6 | **+14*** | **+25*** |
| TimesFM | +15 | +15 | **+25*** | 0 | -3 | **+20*** | **+33*** | **+35*** | **+29*** | **+22*** |
| Chronos-2 | +15 | +15 | **+31*** | +6 | -2 | +18 | **+34*** | **+37*** | **+27*** | +20 |
| DLinear | +10 | +10 | **+39*** | **+27*** | **+27*** | **+25*** | **+19*** | **+42*** | +12 | **+16*** |
| Linear-context | +9 | +9 | **+41*** | **+27*** | **+26*** | **+26*** | **+21*** | **+42*** | +12 | **+16*** |
| 4-week profile | **+31*** | **+31*** | **+25*** | +21 | +5 | **+32*** | +3 | **+43*** | **+44*** | +15 |
| LSTM-T-KAN-M | +12 | +12 | **+60*** | **+38*** | **+37*** | **+20*** | **+19*** | **+53*** | **+24*** | +14 |
| Holt–Winters | **+39*** | **+39*** | **+39*** | +20 | **+37*** | **+39*** | **+39*** | **+50*** | **+38*** | **+41*** |
| LSTM | **+19*** | **+19*** | **+47*** | **+34*** | **+37*** | **+20*** | **+17*** | **+54*** | +23 | **+28*** |
| Previous-month profile | +4 | +4 | **+27*** | **+23*** | +5 | **+35*** | +1 | **+47*** | **+44*** | +17 |
| MSTL | +26 | +26 | **+40*** | +5 | +20 | **+34*** | **+40*** | **+51*** | **+36*** | **+43*** |
| Seasonal naive (week) | +21 | +21 | **+32*** | +7 | 0 | +16 | +7 | **+45*** | **+30*** | **+24*** |
| Last-year profile | **+19*** | **+19*** | **+55*** | **+49*** | **+58*** | +21 | **+24*** | **+32*** | **+35*** | **+37*** |
| GBT (zero-shot) | **+39*** | **+39*** | **+29*** | +19 | +13 | **+42*** | **+44*** | **+49*** | **+38*** | **+26*** |
| Seasonal naive (day) | **+45*** | **+45*** | **+52*** | **+22*** | +19 | **+31*** | **+34*** | **+61*** | **+54*** | **+56*** |

**b Head-to-head on energy**




Same information · trained
Same information · zero-shot foundation model
Load and calendar · trained
Load only · zero-shot foundation model
Load only · statistical
Load only · trained
Per-unit regression (calendar, temperature)
Profiles, naive, zero-shot GBT